Problem Set-2 - Solutions

1] $3.248 \times 10^4 = 32480 = 111111011100000_2$

$= 1.111110111000000 \times 2^{14}$

Bias: $127 + 14 = 141 = 10001101_2$

IEEE Single precision FP Number:

Ans: 0 10001101 11111011100000

2] 0 10011000 10000100010100110000000

10011000 → 152 − 127 = 25 ↳ Decimal!

$1.00000100010100110000000 \times 2^{25}$

$= 100000100010100110000000$

$= \cancel{99411} = \cancel{100000100010100110 \times 2^{10}}$ $= 34121216$

$= \cancel{49705.1 \times 10^3}$ //

3] (a) 01111100001010 11

Sign = 0

$1.111100001010 11 \times 2^{14}$

$e = 127 + 14 = 141 = 10001101$

Single precision notation:

01000110 1111000010101100000000

(b) 10011000001 1000 sign = 1.

$1.10000011000 \times 2^{11}$

$e = 11 + 127 = 138 = 10001010$

Format:

1 10001010 10000011000000000000000

4(a) 1 10000001 01001001110001000000000

$S = 1.$

$e = 10000001 = 129 - 127 = 2.$

$M = \cancel{1.001001110001}\ 1.01001001110001 \times 2^2$

$= \cancel{100.1001110001}\ 101.001001110001 = \cancel{4.6103515625}\ 5.15258789.$

Ans: -4.6103515625

(b) 0 11001100 10000111110100100000000

$S = 0$

$e = 11001100 = 204 - 127 = 77.$

$M = 1.10000111110100 1$

$1.10000111110100 1 \times 2^{77}$

↳ Ans. may be left as it is owing to large decimal value.

(c)
$$\begin{array}{r} 574.6 \\ -279.7 \\ \hline 294.9 \end{array} \Rightarrow \begin{array}{r} 574.6 \\ 720.3 \\ \hline 1294.9 \end{array}$$

Ans: 294.9

(d)
$$\begin{array}{r} 376.3 \\ -765.6 \\ \hline -389.3 \end{array} \Rightarrow \begin{array}{r} 376.3 \\ 234.4 \\ \hline 610.7 \end{array}$$

10's

Ans: −389.3.

6) a) 37 = 00000100101

−37 = 1111111011011 (2's).

b) 173 = 0000101011 01

−173 = 1111010100 11 (2's)

(c) 65.5 = 01000001.1000

−65.5 = 10111110.1000 (2's)

(d) 197.5 = 000011000101.1000

−197.5 = 1111001110 10.1000 (2's)

7) a) 97 = 000001100001

−97 = 111110011110 (1's).

(b) +224 = 000011100000

−224 = 111100011111 (1's.

5 ~~#~~ $\boxed{\text{9's comp}}$

(a)
$$\begin{array}{r} 274 \\ -86 \\ \hline 188 \end{array} \xrightarrow{9's} \begin{array}{r} 274 \\ +913 \\ \hline 1187 \\ +1 \\ \hline 188 \end{array}$$

(b)
$$\begin{array}{r} 93 \\ -615 \\ \hline -522 \end{array} \xrightarrow{9's} \begin{array}{r} 93 \\ 384 \\ \hline 477 \end{array}$$

Ans $477 \xrightarrow{9's} 522$

Ans: -522

c)
$$\begin{array}{r} 574.6 \\ -279.7 \\ \hline 294.9 \end{array} \Rightarrow \begin{array}{r} 574.6 \\ +720.2 \\ \hline 1294.8 \\ 1 \\ \hline 294.9 \end{array}$$

(d)
$$\begin{array}{r} 376.3 \\ -765.6 \end{array} \Rightarrow \begin{array}{r} 376.3 \\ 234.3 \\ \hline 610.6 \end{array}$$

$\downarrow$ 9's

Ans: -389.3

$\boxed{\text{10's comp}}$

(a)
$$\begin{array}{r} 274 \\ -86 \end{array} \xrightarrow{10's} \begin{array}{r} 274 \\ 914 \\ \hline 1188 \end{array}$$

Ans: 188

(b)
$$\begin{array}{r} 93 \\ -615 \\ \hline -522 \end{array} \Rightarrow \begin{array}{r} 93 \\ 385 \\ \hline 478 \end{array}$$

$\downarrow$ 10's.

Ans: -522

(d) 36.75
−89.5 ⇒
−52.75

00100100.1100
10100110.1000 (2's)

11001011.0100

↓ 2's.

00110100.1100 = −52.75

9) (a) 52
−17 ⇒
35

00000110100
11111101110

1 00000100010
1

00000100011 = 35

(b) 46
−84 ⇒
−38

00000101110.
11111010101 1

01111 011001 is → 00000100110 = 38

Ans: −38

(c) Similarly. Ans: 46.25

Soln:
63.75
−17.5
46.25

⇒ 00111111.1100
11101110.0111

00101110.0011
+1

(c) $205.75 = 00001100110 1.1100$

$-205.75 = 11110011001 0.0011$ (1's)

d) $29.375 = 00011101.0110$

$-29.375 = 11100010.1001$ (1's)

8) (a)

$$\begin{array}{r} 46 \\ -19 \\ \hline +27 \end{array} \Rightarrow \begin{array}{l} 00000101110. \\ \underline{11111110110 1} \quad (2's) \\ \cancel{1}\ 00000011011 \end{array}$$

Ans: $11011 = 27$

(b)

$$\begin{array}{r} 27 \\ -75 \end{array} \Rightarrow \begin{array}{l} 00000011011 \\ \underline{11111011010 1} \quad (2's) \\ 11111101 0000 \end{array}$$

$\downarrow$ (2's).

Ans: $00000110000 = -48$

(c)

$$\begin{array}{r} 125.25 \\ -46.75 \end{array} \Rightarrow \begin{array}{l} 00000111101.0100. \\ \underline{11111101 0001.0100} \quad (2's) \\ \textcircled{1}\ 00000100111 0.1000 \end{array}$$

$1011 10.1 = 78.5$

Ans: 00101110.0100 = 46.25.

(d)

$$\begin{array}{r} 73.5 \\ -112.75 \\ \hline -39.25 \end{array} \Rightarrow \begin{array}{r} 01001001.1000 \\ 10001111.0011 \\ \hline 11011000.1011 \end{array}$$

$\rightarrow$ is $\rightarrow$ 00100111.0100

Ans: -39.25

10)

	Signed	1's	2's
a) +14	00001110	00001110	00001110
b) 27	00101101	$\rightarrow$	$\rightarrow$
c) 45	00101101	$\rightarrow$	$\rightarrow$
d) -17	10010001	11101110	11101111
e) -37	10100101	11011010	11011011
f) -76	11001100	10110011	10110100